

DIGITAL TEMPERATURE CONTROLLER USING THERMOCOUPLE



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Digital temperature controllers are essential for temperature measurement and control of instrumentation in industries. These are used for home appliances too. Several types and makes of analogue and digital temperature controllers are available in the market but they are very costly and complex in design.

Temperature measurement basics

There are two types of temperature measurement systems:

1. Direct temperature measurement system, for up to 999°C
2. Indirect temperature measurement system, for a higher temperature range at which physical sensors may burn

Direct temperature measurement

The selection of a temperature controller is dependent on the range of temperature and temperature sensor—the most important and critical part of any temperature measurement and control system.

There are many types of sensors for different ranges of temperature measurement:

1. For temperature range of 0-100°C, the most common sensor is a thermistor or semiconductor sensor.

2. For temperature range of 0-250°C, the most common sensor is RTD PT-100.

3. For temperatures up to 1000°C or more, thermocouples are used as sensors.

Here we are concerned with the direct method of

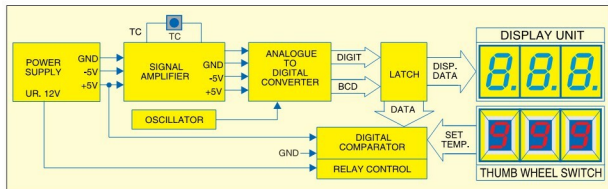


Fig. 1: Block diagram of digital temperature controller

Here is a digital temperature controller that is highly reliable and accurate yet very easy and inexpensive to make. It can measure temperatures in the range of 0 to 999°C.

Salient features of this digital temperature controller are:

1. Universal signal conditioning for any thermocouple input
2. Flexible design for any temperature range
3. Highly accurate and reliable due to 4½-digit analogue-to-digital converter (ADC) ICL7135
4. Easy temperature setting using digital thumbwheel switch
5. Suitable for home and industries
6. No programming or any other software required

system is normally used to measure the temperature of a flame, hot air/gas, or hot body of a metallic surface, in the form of some electrical unit like microvolt (μV) or milliampere (mA), or resistance change as input signal to the measuring instrument.

Indirect temperature measurement system is used where direct temperature measurement is not possible as the temperature sensor may burn due to high temperature or there is no physical contact with the sensor for measurement. In such cases, an indirect temperature measurement technique called 'optical pyrometry' is used. Pyrometer is used for indirect temperature measurement through UV rays and photo-sensors.

measurement for temperatures up to 1000°C, using K-type of thermocouple as the input temperature sensor.

A thermocouple forms when two different metals join together to make a junction. When heated, this junction of two different metals produces electrical energy in microvolts. In other words, heat energy is converted into electrical energy due to thermal effect of materials joined together at a junction.

In K-type thermocouples, the combination of metals like chromel (Ni-Cr) and alumel (Ni-Al) forms the junction. K-type is low-cost and one of the most popular general-purpose thermocouples. Its operating range is around -180°C to +1350°C. Sensitivity is ap-